SOLARIS X – Grand Hackathon

Track: Autonomous Analytics Systems

This track of the hackathon focuses on building intelligent, self-driven systems that can transform raw data into meaningful insights with minimal human intervention. In today's data-driven world, organizations generate massive volumes of data, but the process of converting this data into actionable insights remains complex, time-consuming, and largely manual.

The analytics lifecycle involves multiple stages such as data collection, cleaning, exploratory data analysis, feature engineering, and predictive modeling. These stages often require continuous human effort, leading to inefficiencies, delays, and limited real-time decision-making capabilities.

The goal of this track is to design and develop end-to-end agentic AI pipelines that automate the complete analytics process. Participants are expected to build systems composed of multiple intelligent agents, where each agent is responsible for a specific task, collaboratively working to process data and generate insights autonomously.

These systems should not only analyze data but also understand and present it effectively. For instance, in a mineral extraction use case, the system can process geographical data, identify mineral types, estimate their distribution and quantity, and further predict resource requirements such as manpower, time, and cost. The system can also extend into financial analysis by estimating investment and potential returns.

The format is inspired by next-generation AI systems where automation, intelligence, and collaboration between agents replace traditional step-by-step manual analytics workflows. The focus is on building scalable, efficient, and intelligent systems capable of real-world impact.

The end takeaway is that innovation, automation, and intelligent decision-making win.

What you can build?

This will be an open innovation track with no fixed subtracks, allowing participants to explore diverse domains and develop impactful autonomous analytics solutions.

Approaches that can be taken:

- **Geospatial Analytics:** Systems that analyze maps, satellite data, and terrain information for insights such as mineral detection or land usage.
- **Business Intelligence:** Automated pipelines that generate dashboards, KPIs, and business insights from raw organizational data.

● **Finance:** Predictive systems for investment analysis, cost estimation, and financial forecasting.
● **Healthcare Analytics:** Systems that process medical data to generate insights for diagnosis and treatment planning.
● **Data Engineering:** Intelligent agents that automate data cleaning, transformation, validation, and pipeline orchestration.
● **Smart Cities:** Data-driven systems for traffic analysis, resource allocation, and urban planning.
● **E-Commerce Analytics:** Automated recommendation systems, demand forecasting, and customer behavior analysis.
● **Environmental Analysis:** Systems for climate data analysis, resource monitoring, and sustainability insights.
● **Decision Support Systems:** AI pipelines that assist in strategic decision-making using data-driven insights.
● **Interactive Dashboards:** Visualization systems using tools like Power BI or Tableau for real-time data exploration.

Do's and Don'ts

Do's

● Build complete end-to-end agentic pipelines that automate the analytics lifecycle.
● Focus on real-world datasets and impactful problem statements.
● Ensure clear coordination between multiple intelligent agents.
● Design interactive dashboards for effective visualization and decision-making.

Don'ts

● Don't build isolated models without pipeline integration.
● Don't rely heavily on manual intervention in the workflow.
● Don't create static outputs without meaningful insights or visualization.
● Don't choose overly simple problems with limited real-world applicability.